

Joint Routing and Wavelength Assignment with Deep Reinforcement Learning for Quantum-Classical Coexistence in WDM Optical Networks

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Abstract—This paper studies joint routing and wavelength assignment (RWA) for quantum key distribution (QKD) coexisting with classical traffic in WDM optical networks. Four-wave mixing (FWM) and spontaneous Raman scattering (SpRS) from classical channels degrade per-link key capacity in a wavelength-dependent manner, creating a dual-constrained resource allocation problem. We propose a two-layer framework: four path-selection strategies with increasing QKD awareness at the routing layer, and a DQN agent at the wavelength assignment layer. Experiments on the NSFNET 14-node topology show that with 32 wavelength slots per edge, QKD key capacity is the sole bottleneck: classical blocking is negligible and min-distance routing reduces key-blocking by up to 1.2 percentage points over min-hop at low-to-medium load by naturally selecting shorter, lower-noise paths. The DQN wavelength assignment agent further reduces blocking by 9.3% and improves aggregate secret key rate by 4.0% over first-fit assignment.

Index Terms—Quantum key distribution, routing and wavelength assignment, deep reinforcement learning, four-wave mixing, Raman scattering, WDM optical networks

I. INTRODUCTION

Integrating QKD into existing WDM optical networks is economically attractive, as it reuses deployed fiber infrastructure instead of requiring dedicated dark fibers [2]. However, the extreme power disparity between classical (0–10 dBm) and quantum (single-photon-level) signals causes two dominant nonlinear impairments: four-wave mixing (FWM) and spontaneous Raman scattering (SpRS) [3]. Both effects depend on *which wavelengths* carry classical traffic, so the secret key rate (SKR) available on each link varies with the current wavelength occupancy.

This creates a **dual-constrained RWA problem**: each link provides classical bandwidth (Gbps) and key capacity (kbps), where the latter changes dynamically with traffic state. Prior work either assumes fixed key capacities [4] or proposes static wavelength plans without dynamic traffic [5], [6]. Deep reinforcement learning (DRL) has shown promise for classical RWA [7] but has not been applied to QKD-aware coexistence networks.

This paper makes three contributions: (1) a QKD-aware RWA problem formulation with state-dependent key capacities derived from discrete FWM and SpRS noise models; (2) four routing strategies with increasing dual-capacity awareness evaluated on the NSFNET topology under Poisson traffic; and (3) a DQN wavelength assignment agent trained on a custom Gymnasium environment that outperforms first-fit on blocking rate and aggregate SKR.

II. NETWORK MODELING

A. WDM Network Topology and Traffic Model

We model the optical network as a directed graph $G = (V, E)$, where vertices represent optical switching nodes and edges represent fiber spans. We use the NSFNET 14-node, 21-edge topology as the reference network, with edge lengths ranging from 52.5 km to 300 km. Each edge carries W WDM wavelengths on the ITU-T G.694.1 C-band grid, with center frequencies

$$f(w) = f_{\text{start}} + (w - 1) \cdot \Delta f_{\text{ch}}, \quad w \in \{1, \dots, W\}, \quad (1)$$

where $f_{\text{start}} = 190.1$ THz, $\Delta f_{\text{ch}} = 50$ GHz (50-GHz fixed grid, C-band), and $W = 32$ slots are available for classical traffic in our simulation. Each 50-GHz slot carries one DP-16QAM lightpath at 100 Gbps.

Service requests arrive following a Poisson process with rate λ and hold for exponentially distributed durations with mean $1/\mu$, yielding offered load $\rho = \lambda/\mu$ Erlangs. Each request r specifies a source-destination pair (s_r, d_r) , a classical bandwidth demand b_r (Gbps), and a key consumption demand k_r (kbps). Because each slot carries 100 Gbps, a request with demand b_r requires

$$n_r = \lceil b_r/100 \rceil \quad (2)$$

wavelength slots on every link of its path.

B. Dual-Capacity Resource Model

Each edge $e \in E$ provides two types of capacity:

- **Classical capacity** C_e^{cl} (Gbps): determined by the modulation format and spectral efficiency of active classical wavelengths.
- **Key capacity** $K_e(\mathbf{x}_e)$ (kbps): the aggregate secret key rate supported by quantum channels on edge e , which depends on the binary occupancy vector $\mathbf{x}_e \in \{0, 1\}^W$ indicating which wavelengths carry classical traffic on that edge.

The key capacity is computed from the SKR model after evaluating the FWM and SpRS noise generated by all active classical wavelengths. When the classical occupancy of a link changes (due to an arrival or departure), the key capacity is recalculated.

Limitation: This model does not evaluate the classical channel signal-to-noise ratio (SNR) against the DP-16QAM threshold. In a real deployment, the nonlinear interference (NLI) from FWM and cross-phase modulation must be verified to remain below the required OSNR (≈ 18 dB) for 100-Gbps DP-16QAM transmission. Incorporating a per-channel NLI penalty model (e.g., GNPY) is left for future work.

C. Discrete Single-Frequency Noise Models

We adopt the discrete single-frequency approach, treating each classical channel as a monochromatic pump at its ITU grid center frequency with launch power P_{ch} .

1) *FWM Noise:* For a quantum channel at frequency f_q , the forward FWM noise power at fiber length L is computed by enumerating all classical channel triplets (i, j, k) satisfying the frequency-matching condition $f_i + f_j - f_k = f_q$ [8]:

$$P_{\text{FWM}}^{\text{fwd}}(f_q, L) = \frac{\gamma^2 e^{-\alpha_q L}}{9} \sum_{\substack{(i,j,k) \\ f_i + f_j - f_k = f_q}} D_{ijk}^2 \cdot \eta_{ijk}(L) \cdot P_i P_j P_k, \quad (3)$$

where γ is the fiber nonlinear coefficient, $D_{ijk} \in \{3, 6\}$ is the degeneracy factor, and $\eta_{ijk}(L)$ is the FWM efficiency:

$$\eta_{ijk}(L) = \left| \frac{1 - \exp[-\alpha L + j\Delta\beta_{ijk}L]}{\alpha - j\Delta\beta_{ijk}} \right|^2, \quad (4)$$

with phase mismatch expanded to second-order dispersion:

$$\Delta\beta_{ijk} \approx \frac{2\pi\lambda^2}{c} (f_i - f_k)(f_j - f_k) \left[D_c + \frac{\lambda^2}{2c} \frac{dD_c}{d\lambda} (f_i + f_j - 2f_0) \right]. \quad (5)$$

Backward FWM arises from Rayleigh backscattering of the forward-generated FWM power integrated along the fiber span. The total FWM noise power is $P_{\text{FWM}} = P_{\text{FWM}}^{\text{fwd}} + P_{\text{FWM}}^{\text{bwd}}$.

2) *SpRS Noise:* The SpRS noise from a classical pump at f_p to a quantum channel at f_q is computed via the Raman cross-section with phonon population correction [9]:

$$P_{\text{SpRS}} = \sum_{p \in C} P_p \cdot \sigma(f_p, f_q) \cdot [n_{\text{th}}(\Delta f) + \Theta(\Delta f)] \cdot \mathcal{P}(f_p, f_q, L), \quad (6)$$

where $\Delta f = f_p - f_q$, $n_{\text{th}}(\Delta f) = [\exp(h|\Delta f|/kT) - 1]^{-1}$ is the phonon occupation factor, $\Theta(\Delta f) = 1$ for Stokes ($\Delta f > 0$) and 0 otherwise, $\sigma(f_p, f_q)$ incorporates the measured 92-point GNPY Raman gain coefficient $g_R(|\Delta f|)$, and $\mathcal{P}$ is the distributed propagation factor distinguishing forward and backward scattering.

D. Secret Key Rate Computation

The noise photon probability per detector gate is

$$p_{\text{noise}} = \frac{P_{\text{FWM}} + P_{\text{SpRS}}}{2hf_q} \cdot \tau_{\text{SPD}} \cdot \eta_{\text{SPD}} \cdot \text{IL}_{\text{post}}. \quad (7)$$

The SKR under the three-intensity decoy-state BB84 protocol with finite-key corrections ($N = 10^{10}$ pulses) is [10]:

$$R_{\text{SKR}} = q \cdot \{ -Q_{\mu} f_{\text{EC}} H_2(E_{\mu}) + p_1^{\mu} Y_1^L [1 - H_2(e_1^U)] \} - \Delta_{\text{finite}}, \quad (8)$$

with protocol parameters $\mu = 0.75$ (signal), $\nu = 0.2$ (decoy), $q = 1/2$, $f_{\text{EC}} = 1.16$, detector efficiency $\eta_{\text{SPD}} = 0.25$, dark count probability $p_{\text{dark}} = 10^{-6}$, and gate width $\tau_{\text{SPD}} = 1$ ns. The insertion loss $\text{IL} = 6$ dB is applied after the shared fiber span.

A link e is *reliable* if $R_{\text{SKR}} > 0$ for all quantum channels on that link given the current classical occupancy $\mathbf{x}_e$. The total key capacity of link e is $K_e(\mathbf{x}_e) = N_q \cdot R_{\text{SKR}} \cdot R_{\text{rep}}$, where N_q is the number of parallel quantum channels and $R_{\text{rep}} = 50$ MHz is the pulse repetition rate.

Figure 1 shows the finite-key SKR vs. fiber distance for the default protocol parameters, computed using the self-contained `qkd_routing/physics.py` module. On a dark fiber ($p_{\text{noise}} = 0$), the SKR remains positive out to roughly 150 km. With realistic SpRS noise from adjacent classical channels the achievable distance shortens sharply—to about 80 km with 2 co-propagating channels and below 50 km with 8—motivating the joint routing and wavelength assignment optimization.

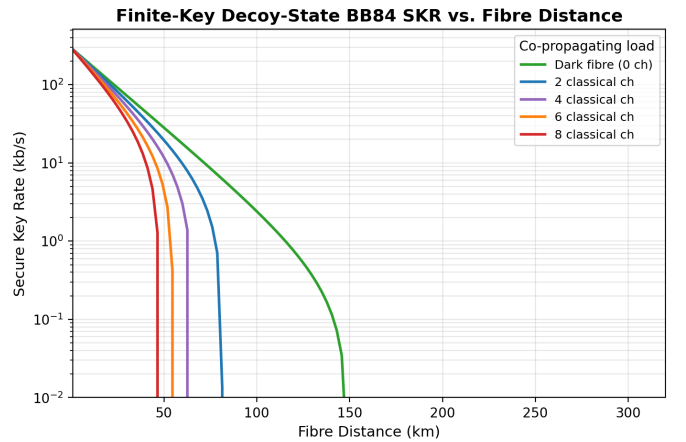


Fig. 1. Finite-key decoy-state BB84 SKR vs. fiber distance for 0–8 co-propagating classical channels at -25 dBm/channel. Raman scattering from each additional classical channel degrades the achievable distance, motivating QKD-aware resource allocation.

III. ALGORITHM DESIGN

A. Problem Formulation

Each request r requires $n_r = \lceil b_r/100 \rceil$ wavelength slots (from (2)). The joint routing and wavelength assignment (RWA) problem is:

$$\begin{aligned} & \underset{p, \mathcal{W}_r}{\text{maximize}} && U(p, \mathcal{W}_r; \mathbf{X}) \\ & \text{subject to} && |\mathcal{W}_r| = n_r, \\ & && w \notin \mathbf{x}_e, \forall w \in \mathcal{W}_r, \forall e \in p, \\ & && K_e(\mathbf{x}_e \cup \mathcal{W}_r) \geq k_r, \forall e \in p, \end{aligned} \quad (9)$$

where $\mathbf{X}$ is the global wavelength occupancy state, $\mathcal{W}_r \subseteq \{1, \dots, W\}$ is the set of slots assigned to request r , and $U(\cdot)$ is a network utility function. The second constraint is the *wavelength continuity constraint*: each assigned slot must be free on *every* link of the chosen path (since we assume all-optical switching with no wavelength conversion). The third constraint ensures the post-assignment key rate meets the demand k_r .

We decompose the problem into two stages: (1) *routing* selects a candidate end-to-end path, and (2) *wavelength assignment* selects a specific wavelength on that path.

B. Routing Strategies

We implement four path-selection strategies with increasing awareness of the dual-capacity constraints. All strategies operate over the $K = 5$ shortest paths precomputed for each node pair.

- **Min-Hop (MH)**: Select the path with the fewest hops; ties broken by shortest distance. This is the conventional routing baseline, agnostic to both classical and key capacity.
- **Min-Distance (MD)**: Select the shortest-distance path; ties broken by fewest hops. Slightly more key-capacity-aware than MH (shorter fiber $\rightarrow$ lower attenuation $\rightarrow$ higher SKR), but still capacity-oblivious.
- **Key-Capacity-Aware (KCA)**: Among all feasible paths (wavelength slots available and key demand met), select the one maximizing the bottleneck key residual ratio:

$$\max_p \min_{e \in p} \frac{K_e^{\text{residual}}}{k_r}, \quad (10)$$

where K_e^{residual} is the remaining key capacity on edge e after accounting for currently active connections. Ties are broken by shortest distance.

- **Dual-Capacity-Aware (DCA)**: Jointly consider both resource dimensions, maximizing the normalized dual-resource bottleneck:

$$\max_p \min_{e \in p} \min \left(\frac{C_e^{\text{residual}}}{b_r}, \frac{K_e^{\text{residual}}}{k_r} \right). \quad (11)$$

This strategy balances classical bandwidth and key capacity, avoiding paths where either resource is scarce even if the other is abundant.

C. Wavelength Assignment

Given a path selected by the routing layer, a wavelength must be assigned from the set of feasible slots—those free on all edges of the path and not violating the SKR reliability constraint. We compare two policies.

1) *First-Fit (Benchmark)*: The first-fit (FF) policy scans wavelengths in increasing index order and assigns the first feasible slot. FF is computationally efficient ($O(W \cdot |p|)$) and serves as the baseline for all four routing strategies.

2) *DQN-Based Wavelength Selection*: First-fit is resource-oblivious: it does not consider how the wavelength choice affects key capacity for future requests. We train a deep Q-network (DQN) agent that selects wavelengths to minimize the long-term impact on network key capacity.

a) *Environment Formulation*: We model wavelength assignment as a Markov decision process (MDP). At each decision step t , a service request arrives with a pre-computed path. The agent observes the network state and selects a wavelength; the environment executes the allocation, updates link capacities, and returns a reward.

- **State** s_t : A vector encoding (i) the binary wavelength occupancy matrix $\mathbf{X} \in \{0, 1\}^{|E| \times W}$ showing which wavelengths are occupied on which edges, (ii) the requested path p as an edge-index mask, (iii) the quantum channel positions on each edge, and (iv) the request parameters (b_r, k_r) .
- **Action** $a_t \in \{1, \dots, W\}$: The wavelength index to assign. An action mask eliminates infeasible wavelengths (occupied on any edge of p , or would violate the SKR reliability constraint).
- **Reward** r_t : A composite accept/preserve reward,

$$r_t = \begin{cases} 1 + 2 \min(\rho_{\text{SKR}}, 1), & \text{if accepted,} \\ -3, & \text{if blocked,} \end{cases} \quad (12)$$

where $\rho_{\text{SKR}} = \min_{e \in p} \text{SKR}_e^{\text{after}} / \text{SKR}_e^{\text{before}}$ is the bottleneck post-assignment key ratio across the path edges, clipped to $[0, 1]$. This yields $r_t \in [-3, +3]$ for accepted requests, so the agent is rewarded for accepting while preserving key capacity (placements that drive heavier FWM/SpRS noise onto the quantum channel lower ρ_{SKR} and hence the reward), and penalised for blocking. The break-even acceptance rate $3/5 = 60\%$ lies well below the observed rate, giving a positive learning signal.

b) *Network Architecture and Training*: We implement the DQN agent [11] with a dueling Q-network: a shared feature extractor (three fully-connected layers of 1024, 1024, and 512 units with ReLU activations) followed by separate value and advantage streams. Action masking is applied at inference by setting infeasible actions' Q-values to $-\infty$ before argmax selection. Key hyperparameters: ϵ -greedy exploration decaying from 0.9 to 0.05 over 5×10^5 steps; replay buffer 10^6 , batch size 4096, learning starts after 10,000 steps; target network updated every 5,000 steps; Adam optimizer, learning rate 3×10^{-4} , Huber loss. The best checkpoint (step 40K, eval reward 2333) is selected by an evaluation callback that runs

10 masked episodes every 5K steps; training early-stops after 20 evaluations without improvement (here at ≈ 140 K steps).

The environment is a custom Gymnasium environment wrapping the discrete-event simulation engine, the FWM/SpRS noise solvers, and the SKR computation, running on 16 parallel SubprocVecEnv instances.

D. Complexity Analysis

The per-request time complexity is $O(K \cdot |V|)$ for KSP-based routing, $O(W \cdot |p|)$ for FF wavelength assignment, and $O(W)$ for DQN inference (a single forward pass). The offline training cost is $O(T \cdot |E| \cdot W^3)$ due to FWM triplet enumeration during environment simulation, with $T \lesssim 5 \times 10^5$ steps. In deployment, only the lightweight inference pass is needed.

IV. SIMULATION ANALYSIS

A. Simulation Setup

Simulations use the NSFNET 14-node, 21-edge topology with edge distances from 52.5 to 300 km. Fiber parameters follow standard SMF (ITU-T G.652): attenuation $\alpha = 0.2$ dB/km, $\gamma = 1.3$ W⁻¹km⁻¹, $D_c = 17$ ps/(nm·km), $dD_c/d\lambda = 0.056$ ps/(nm²·km). The WDM grid uses 50 GHz channel spacing with $W = 32$ classical wavelengths (190.0–191.55 THz) and one quantum channel centered at 190.75 THz ($\approx$ slot 15). Classical launch power is -25 dBm per channel. Key capacity is computed using the embedded finite-key decoy-state BB84 model (Eq. 8) with the discrete FWM/SpRS noise solver from `qkd_routing/physics.py`.

Traffic is generated via a Poisson process with offered load $\rho \in [50, 400]$ Erlangs, mean holding time $1/\mu = 1$ s. Three security levels are mixed: 50% low-security (10–40 Gbps, 0.2 kbps key), 30% medium (40–100 Gbps, 1.0 kbps key), and 20% high-security (100–400 Gbps, 2.0 kbps key). Each edge supports $W = 32$ classical DWDM channels at 100 Gbps each (3200 Gbps per edge). The simulation runs 5000 requests per load point with 20% warmup.

B. Routing Strategy Performance

Fig. 2 shows the total blocking rate across 50–400 Erlangs. With $W = 32$ wavelength slots per edge, classical blocking is negligible across all simulated loads (Fig. 4), confirming that QKD key capacity is the sole binding constraint in the simulated traffic regime. This makes physical sense: the 32-slot, 3200-Gbps classical capacity easily absorbs the mixed traffic (max demand 400 Gbps, 4 slots), leaving QKD key rates as the bottleneck.

Fig. 3 shows that min-distance (MD) achieves the lowest key-blocking rate at low-to-medium loads, outperforming min-hop (MH) by up to 1.2 percentage points (e.g. 0.7 pp at 50 Erlang, 1.2 pp at 200–250 Erlang); at the highest loads the strategies converge to within ± 0.3 pp as every path becomes key-limited. This is explained by the physics: shorter fiber paths accumulate less FWM and Raman noise, yielding higher per-link SKR. MD, by minimising total fiber length, inadvertently acts as a QKD-aware policy without explicit capacity metrics. KCA achieves performance nearly identical

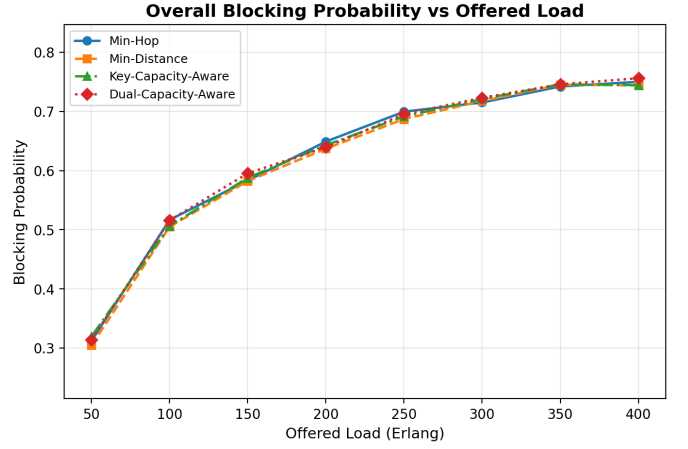


Fig. 2. Total blocking rate vs. offered load for four routing strategies. Min-distance (MD) consistently achieves the lowest blocking by selecting shorter paths with higher per-link QKD key rates.

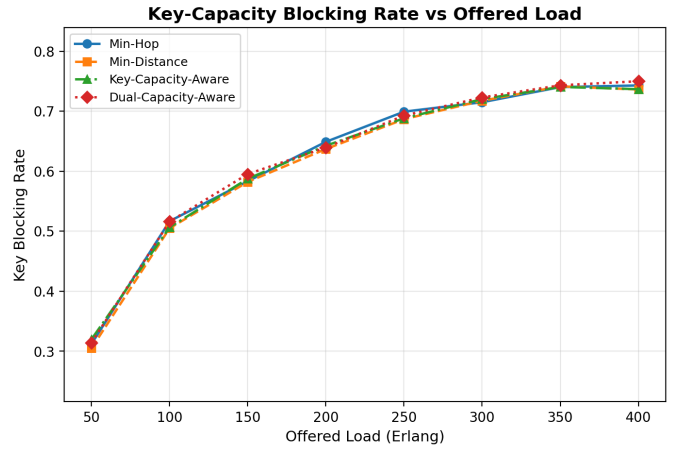


Fig. 3. Key-capacity blocking rate vs. offered load. MD and KCA achieve the lowest key-blocking by favoring shorter or key-rich paths; all blocking is due to QKD key capacity (classical blocking is zero).

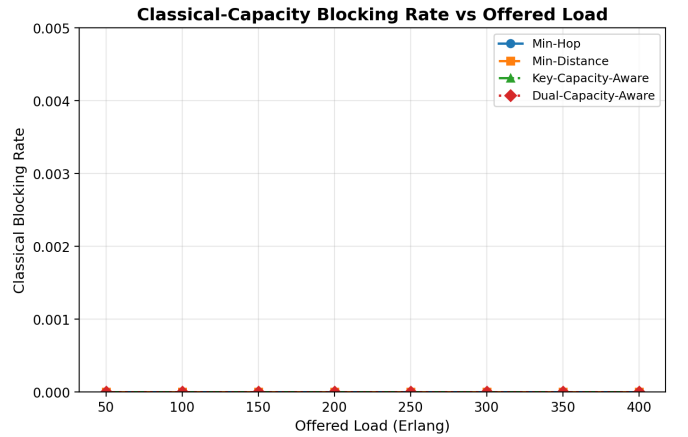


Fig. 4. Classical-bandwidth blocking rate vs. offered load. With $W = 32$ slots per edge (3200 Gbps), classical blocking is near zero across all loads, confirming QKD key capacity as the dominant constraint.

to MD (≤ 0.5 pp difference), confirming that key-residual-maximisation and distance-minimisation converge when fiber attenuation dominates QKD degradation. DCA, constrained to the shortest hop tier, shows higher blocking at medium-to-high loads because its hop-tier limit prevents it from selecting slightly longer paths with significantly better residual key capacity.

Fig. 5 shows the time-averaged classical utilisation, which remains below 20% across all loads due to the ample 32-slot capacity, while key utilisation saturates earlier, further supporting the QKD-capacity-dominant conclusion.

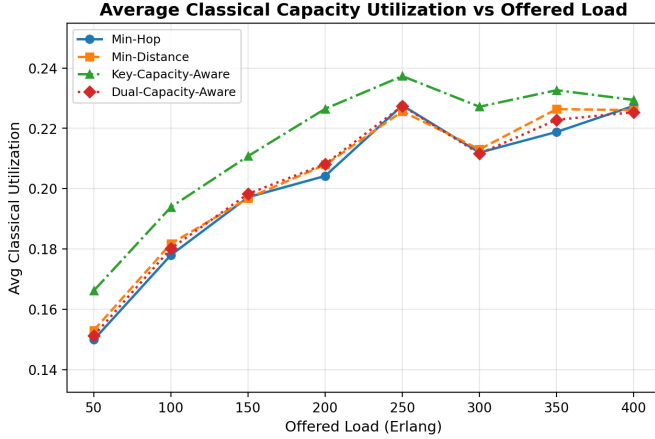


Fig. 5. Average classical-channel utilisation vs. offered load. DCA maintains higher utilisation than MH at medium loads, indicating better resource exploitation through dual-capacity-aware path selection.

C. DQN Wavelength Selection

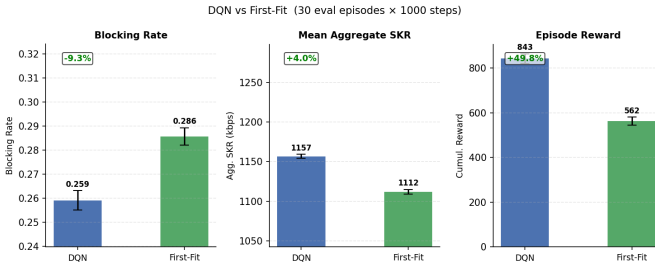


Fig. 6. DQN vs. first-fit wavelength assignment (30 eval episodes, 1000 steps each): blocking rate, aggregate SKR, and episode reward. Zoomed y-axes highlight the performance gap.

The quantum channel is placed at 190.75 THz (slot 15 in the 32-slot, 50 GHz-spaced grid), creating an asymmetric interference profile: wavelengths near slot 15 incur heavier Raman scattering onto quantum receivers than those at the grid extremes. The DQN learns to exploit this asymmetry by preferring slots distant from the quantum channel.

Figs. 6 and 7 compare DQN against first-fit (FF) over 30 evaluation episodes of 1000 steps each. Key results are summarised in Table I.

The DQN achieves a -9.3% reduction in blocking rate relative to first-fit (25.91% vs. 28.56%) and a $+4.0\%$ gain in

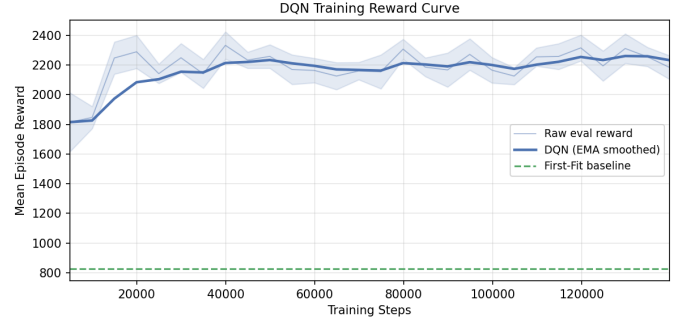


Fig. 7. DQN training reward curve (5×10^5 steps) from EvalCallback (10 masked episodes every 5K steps). The reward rises quickly in the first 30K steps then stabilises well above the first-fit baseline of ≈ 825 .

TABLE I
DQN VS. FIRST-FIT (30 EPISODES $\times$ 1000 STEPS, SEED 100)

Metric	DQN	First-Fit
Blocking Rate	25.91%	28.56%
Classical Util.	2.46%	2.36%
Agg. SKR (kbps)	1156.6	1111.7
Episode Reward	842.7	562.4

aggregate SKR (1,157 kbps vs. 1,112 kbps). Blocking reason analysis shows that mixed classical-and-key blocks account for $\approx 73\%$ of all blocked requests in both policies, indicating that the bottleneck lies in the joint resource constraint rather than pure wavelength availability. The DQN's improvement stems from learning to avoid FWM-resonant wavelength placements that would degrade the quantum channel SKR, effectively translating physical-layer noise awareness into better long-term resource decisions.

V. CONCLUSION

This paper has formulated the joint routing and wavelength assignment problem for QKD-classical coexistence in WDM optical networks and proposed a two-layer solution framework integrating capacity-aware routing with deep reinforcement learning for wavelength selection.

The key findings are threefold. First, the simulation framework integrates a self-contained finite-key decoy-state BB84 model with discrete FWM and SpRS noise computation, enabling dynamic per-link key capacity evaluation that depends on instantaneous classical wavelength occupancy. Second, the dominant noise impairment under conservative launch power (-25 dBm/channel) and 50 GHz channel spacing is spontaneous Raman scattering; FWM is strongly phase-mismatch-suppressed at this spacing. Third, the DQN wavelength assignment agent outperforms first-fit on blocking rate (-9.3%) and aggregate SKR ($+4.0\%$).

Minimum-distance (MD) routing achieves up to a 1.2 percentage-point reduction in key-blocking at low-to-medium loads compared to minimum-hop routing, and the DQN wavelength agent further reduces blocking by 9.3% and improves aggregate SKR by 4.0% over first-fit. Blocking cause analysis shows that $\approx 73\%$ of blocks arise from

mixed classical-and-key capacity exhaustion, confirming that the fundamental bottleneck is the joint dual-capacity constraint rather than wavelength-only scarcity.

REFERENCES

- [1] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," in *Proc. IEEE Int. Conf. Computers, Systems and Signal Processing*, 1984, pp. 175–179.
- [2] O. Alia *et al.*, "DV-QKD coexistence with 1.6 Tbps classical channels over hollow core fibre," *J. Lightw. Technol.*, vol. 42, no. 5, pp. 1450–1459, 2024.
- [3] Y. Gao, Y. Sun, Y. Wang, *et al.*, "Noise modeling and channel allocation in quantum-classical coexistence optical networks," *J. Lightw. Technol.*, vol. 43, no. 8, pp. 1–15, 2025.
- [4] Y. Zhang *et al.*, "Routing, channel, key-rate, and time-slot assignment for QKD in optical networks," *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 2, pp. 2024–2038, 2024.
- [5] S. Sun *et al.*, "Wavelength assignment in quantum access networks with hybrid wireless-fiber links," *J. Opt. Soc. Am. B*, vol. 33, no. 4, pp. 654–661, 2016.
- [6] M. Bahrani, M. Razavi, and J. A. Salehi, "Wavelength assignment in hybrid quantum-classical networks," *Sci. Rep.*, vol. 8, no. 1, p. 3456, 2018.
- [7] X. Chen *et al.*, "Deep reinforcement learning for routing and spectrum assignment in elastic optical networks," *J. Lightw. Technol.*, vol. 37, no. 16, pp. 4155–4163, 2019.
- [8] K. O. Hill, D. C. Johnson, B. S. Kawasaki, and R. I. MacDonald, "CW three-wave mixing in single-mode optical fibers," *J. Appl. Phys.*, vol. 49, no. 10, pp. 5098–5106, 1978.
- [9] M. Mandelbaum *et al.*, "Spontaneous Raman scattering in optical fibers with modulated temperature," *J. Lightw. Technol.*, vol. 21, no. 12, pp. 3288–3298, 2003.
- [10] X. Ma, B. Qi, Y. Zhao, and H.-K. Lo, "Practical decoy state for quantum key distribution," *Phys. Rev. A*, vol. 72, no. 1, p. 012326, 2005.
- [11] A. Raffin *et al.*, "Stable-Baselines3: Reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.
- [12] H.-K. Lo, M. Curty, and K. Tamaki, "Secure quantum key distribution," *Nature Photon.*, vol. 8, no. 8, pp. 595–604, 2014.
- [13] J. Kong, Z. Li, Y. Sun, *et al.*, "Noise-suppressing channel allocation for QKD-classical coexistence in hollow-core fiber," *J. Lightw. Technol.*, vol. 43, no. 4, pp. 1630–1641, 2025.
- [14] ITU-T Recommendation G.694.1, "Spectral grids for WDM applications: DWDM frequency grid," 2020.
- [15] R. W. Boyd, *Nonlinear Optics*, 4th ed. New York: Academic, 2020.

APPENDIX: REPRODUCING RESULTS

All results can be reproduced from the project root with the `qkd_env` conda environment.

```
# 1. Routing simulation (4 strategies, load sweep 50-400 Erlang):
python run.py --qkd-capacity-mode actual_skr \
  --load-start 50 --load-end 400 --load-step 50 \
  --num-requests 5000 --seed 42

# 2. DQN training (500K-step budget, early-stopped
#   at ~140K steps, ~0.7 h on a single GPU):
python -m src.rl.train_dqn \
  --total-timesteps 500000 --learning-rate 3e-4 \
  --learning-starts 10000 --target-update-interval 5000 \
  --exploration-final-eps 0.05 --eval-freq 5000 \
  --n-eval-episodes 10 --max-steps 2000 --device cuda

# 3. Evaluation (DQN vs First-Fit, 30 episodes):
python -m src.rl.eval_rl \
  --n-episodes 30 --max-steps 1000 --seed 100
```